

Easy (10)

1.

What command configures a static route to network 192.168.10.0/24 via next-hop 10.0.0.2?

- A) ip route 192.168.10.0 255.255.255.0 10.0.0.2
- B) route static 192.168.10.0 255.255.255.0 10.0.0.2
- C) static route add 192.168.10.0 255.255.255.0 10.0.0.2
- D) ip static 192.168.10.0 255.255.255.0 10.0.0.2

Answer: A

2.

Which routing protocol uses hop count as its metric?

- A) OSPF
- B) RIP
- C) EIGRP
- D) IGRP

Answer: B

3.

What is the maximum hop count allowed in RIP?

- A) 10
- B) 15
- C) 20
- D) 30

Answer: B

4.

Which command enables RIP routing on a Cisco router?

- A) router eigrp

- B) router rip
- C) router ospf
- D) router igrp

Answer: B

5.

Which routing protocol is Cisco proprietary?

- A) RIP
- B) OSPF
- C) EIGRP
- D) BGP

Answer: C

6.

What does the command no auto-summary do in RIP or EIGRP?

- A) Enables route summarization
- B) Disables automatic route summarization
- C) Resets routing table
- D) Turns off routing

Answer: B

7.

In OSPF configuration, the area 0 is called:

- A) Stub area
- B) Backbone area
- C) Transit area
- D) External area

Answer: B

8.

Which protocol uses the DUAL algorithm?

- A) RIP
- B) IGRP
- C) EIGRP
- D) OSPF

Answer: C

9.

The command to enter router configuration mode for OSPF is:

- A) router ospf [process-id]
- B) router rip
- C) router eigrp [AS]
- D) router igrp [AS]

Answer: A

10.

What type of routing protocol is OSPF?

- A) Distance Vector
- B) Link State
- C) Path Vector
- D) Hybrid

Answer: B

Medium (30)

11.

Which command advertises the network 192.168.1.0/24 in RIP?

- A) network 192.168.1.0
- B) network 192.168.1.0 255.255.255.0
- C) advertise 192.168.1.0
- D) ip route 192.168.1.0 255.255.255.0

Answer: A

12.

How often does RIP send routing updates by default?

- A) 10 seconds
- B) 15 seconds
- C) 30 seconds
- D) 60 seconds

Answer: C

13.

What is the administrative distance of static routes by default?

- A) 0
- B) 1
- C) 90
- D) 120

Answer: B

14.

In EIGRP, what metric components are used for route calculation?

- A) Hop count only
- B) Bandwidth and delay only
- C) Bandwidth, delay, load, reliability

D) Cost and hop count

Answer: C

15.

Which command disables automatic summarization in EIGRP?

A) no auto-summary

B) auto-summary disable

C) summary off

D) no summary

Answer: A

16.

What is the role of the Area Border Router (ABR) in OSPF?

A) Connects different autonomous systems

B) Connects different OSPF areas

C) Manages routing within a single area

D) Connects to external networks

Answer: B

17.

Which routing protocol supports unequal-cost load balancing?

A) RIP

B) OSPF

C) EIGRP

D) IGRP

Answer: C

18.

Which Cisco IOS command shows OSPF neighbors?

- A) show ip ospf neighbor
- B) show ospf neighbors
- C) show ip neighbors ospf
- D) show ospf neighbor

Answer: A

19.

In RIP version 2, what type of message is used to send updates?

- A) Broadcast
- B) Multicast
- C) Unicast
- D) Anycast

Answer: B

20.

Which command enables RIP version 2 on a Cisco router?

- A) version 2 under router rip mode
- B) enable rip v2
- C) ip rip version 2
- D) router rip version 2

Answer: A

21.

Which routing protocol is classful and does not support VLSM?

- A) RIP v1
- B) RIP v2
- C) OSPF
- D) EIGRP

Answer: A

22.

Which of the following is NOT true about IGRP?

- A) It is Cisco proprietary
- B) It uses hop count as the only metric
- C) Supports composite metrics
- D) Has max hop count of 255

Answer: B

23.

What is the wildcard mask for subnet mask 255.255.255.0?

- A) 0.0.0.255
- B) 255.255.255.0
- C) 0.0.255.255
- D) 255.255.0.0

Answer: A

24.

How does OSPF treat a network assigned to area 0?

- A) Backbone area
- B) Stub area
- C) Transit area
- D) External area

Answer: A

25.

Which routing protocol requires a process ID during configuration?

- A) RIP
- B) EIGRP
- C) OSPF

D) IGRP

Answer: C

26.

What is the default administrative distance of RIP?

A) 90

B) 100

C) 120

D) 1

Answer: C

27.

Which command would remove a static route?

A) no ip route [dest] [mask] [next-hop]

B) remove ip route [dest]

C) clear ip route [dest]

D) delete static route [dest]

Answer: A

28.

What algorithm does RIP use for routing updates?

A) Dijkstra's

B) Bellman-Ford

C) DUAL

D) SPF

Answer: B

29.

Which protocol is considered a hybrid routing protocol?

- A) RIP
- B) EIGRP
- C) OSPF
- D) IGRP

Answer: B

30.

Which of the following protocols supports route summarization?

- A) RIP
- B) EIGRP
- C) OSPF
- D) All of the above

Answer: D

31.

In EIGRP, what is a "feasible successor"?

- A) The primary route
- B) Backup route that meets feasibility condition
- C) Route learned from static routing
- D) Route with the highest metric

Answer: B

32.

Which command displays the EIGRP neighbors?

- A) show ip eigrp neighbors
- B) show eigrp neighbors
- C) show ip neighbors eigrp
- D) show neighbors eigrp

Answer: A

33.

OSPF uses which metric to determine the best path?

- A) Hop count
- B) Bandwidth cost
- C) Delay
- D) Load

Answer: B

34.

What is the purpose of the network command under routing protocols?

- A) To define which networks are advertised
- B) To define access control lists
- C) To assign IP addresses
- D) To create VLANs

Answer: A

35.

What is the default administrative distance of EIGRP internal routes?

- A) 90
- B) 110
- C) 120
- D) 100

Answer: A

36.

Which of the following protocols is deprecated?

- A) OSPF
- B) IGRP
- C) EIGRP

D) RIP v2

Answer: B

37.

In OSPF, what is the purpose of a Designated Router (DR)?

A) To reduce routing update traffic on multi-access networks

B) To connect OSPF areas

C) To manage static routes

D) To summarize routes

Answer: A

38.

What does the router eigrp 100 command do?

A) Enables EIGRP for AS number 100

B) Sets router ID to 100

C) Creates 100 EIGRP neighbors

D) Starts EIGRP debugging

Answer: A

39.

Which routing protocol supports authentication of routing updates?

A) RIP v1

B) OSPF

C) IGRP

D) Static routing

Answer: B

40.

Which Cisco command verifies the routing protocols running on a router?

- A) show ip protocols
- B) show running-config
- C) show ip route
- D) show protocols

Answer: A

Hard (10)

41.

What happens when no auto-summary is configured in EIGRP?

- A) It stops summarizing networks at classful boundaries allowing discontinuous subnets
- B) It disables route aggregation entirely
- C) It disables load balancing
- D) It forces summarization on all interfaces

Answer: A

42.

What is the administrative distance of an external OSPF route?

- A) 110
- B) 90
- C) 120
- D) 170

Answer: D

43.

Which metric components does IGRP consider in its composite metric?

- A) Bandwidth, delay, reliability, load, MTU
- B) Hop count, bandwidth, delay
- C) Bandwidth, delay, load, reliability
- D) Bandwidth, delay, MTU, cost

Answer: C

44.

In OSPF, what is the function of an Autonomous System Boundary Router (ASBR)?

- A) Connects OSPF to other routing domains
- B) Summarizes routes between areas
- C) Creates link-state advertisements inside an area
- D) Handles static routes

Answer: A

45.

Which EIGRP feature allows the use of multiple paths with different metrics?

- A) Equal-cost load balancing
- B) Unequal-cost load balancing with variance
- C) Route summarization
- D) Route redistribution

Answer: B

46.

In RIP, what type of messages are used to detect neighbor failure?

- A) Request messages
- B) Acknowledgment messages
- C) Triggered updates
- D) Hello messages

Answer: D

47.

How does OSPF determine the cost of a link by default?

- A) Fixed cost of 1 per link
- B) Based on bandwidth (100 Mbps / link bandwidth in Mbps)
- C) Based on delay in microseconds
- D) Based on hop count

Answer: B

48.

Which command can be used to verify the OSPF database?

- A) show ip ospf database
- B) show ospf neighbors
- C) show ip route ospf
- D) show ip protocols

Answer: A

49.

What is the default hello interval in OSPF on broadcast networks?

- A) 5 seconds
- B) 10 seconds
- C) 30 seconds
- D) 60 seconds

Answer: B

50.

Which command enables EIGRP stub routing on a router?

- A) eigrp stub
- B) router eigrp stub
- C) eigrp stub connected

D) router eigrp 100 stub

Answer: D